CS-2008 Numerical Computing **Quiz 3 - BCS-7A** **Fall 2025**

Name:_____ Roll Number: _____

1. Consider the integral $I = \int_0^2 (x^3 + 1)dx$. What is the approximation of I using the basic **Trapezoidal Rule** ($n = 1$)?

$f(x) = x^3 + 1$

$f(2) = 9$, $f(0) = 1$

$I = \frac{b-a}{2}[f(a) + f(b)]$

$I = \frac{2-0}{2}[1+9]$

$I = 10$

A. 4

B. 10

C. 6

D. 5

2. For the integral $I = \int_0^2 (x^3 + 1)dx$, what is the approximation of I using the basic **Midpoint Rule** ($n = 1$)?

$x_{mid} = \frac{0+2}{2} = 1$

$f(1) = 2$

$I = (b-a) f\left(\frac{a+b}{2}\right)$

$I = 2.2$

$I = 4$

A. 6

B. 10

C. 5.25

D. 4

3. Calculate the approximation of $I = \int_0^2 (x^3 + 1)dx$ using the basic **Simpson's $1/3$ Rule.** (Note: The basic rule requires $n = 2$ intervals).

A. 7.5

B. 6

C. 6.25

D. 5

$$I \approx \frac{h}{3}[f_0 + 4f_1 + f_2]$$

$$\approx \frac{b-a}{6}[1 + 4(2) + 9]$$

$$= \frac{2}{6}[18] = 6$$

4. What is the approximation of $I = \int_0^2 (x^3 + 1)dx$ using the basic **Simpson's $3/8$ Rule?** (Note: The basic rule requires $n = 3$ intervals).

A. 6.125

B. 4.8

C. 5.5

D. 6

$$I = \frac{3h}{8}(f_0 + 3f_1 + 3f_2 + f_3)$$

$$I = 3.\frac{\frac{2}{3}}{8}\left[f_0 + 3f\left(\frac{2}{3}\right) + 3f\left(\frac{4}{3}\right) + f(2)\right]$$

$$I = \frac{2}{8}\left[1 + 3.\frac{35}{27} + 3.\frac{91}{27} + 9\right]$$

$$I = \frac{1}{4} \cdot 24$$

$$I = 6$$

5. Use the **Composite Trapezoidal Rule** with $n = 4$ intervals to approximate $I = \int_0^2 (x^3 + 1)dx$.

A. 6.25

B. 6.0625

C. 6

D. 5.875

$$I = \frac{h}{2}\left[f_0 + 2\sum_{i=1}^{3} f_i + f_4\right]$$

$$I = \frac{b-a}{4.2}\left[f_0 + 2(f_1 + f_2 + f_3) + f_4\right]$$

$$I = \frac{2-0}{8}\left[1 + 2(1.125 + 2 + 4.375) + 9\right]$$

$$I = \ldots$$

$$I = 6.25$$

6. Use the **Composite Midpoint Rule** with $n = 4$ intervals to approximate $I = \int_0^2 (x^3 + 1)dx$.

A. 5.875

B. 5.75

C. 6.25

D. 6

$$h = \frac{b-a}{4} = \frac{2-0}{4} = 0.5$$

$x_0 = 0,\ x_1 = 0.5,\ x_2 = 1,\ x_3 = 1.5,\ x_4 = 2$

$$\bar{x}_1 = \frac{0+0.5}{2} = 0.25$$

$$\bar{x}_2 = \frac{0.5+1}{2} = 0.75$$

$$\bar{x}_3 = \frac{1+1.5}{2} = 1.25$$

$$\bar{x}_4 = \frac{1.5+2}{2} = 1.75$$

$$I = \sum_{i=1}^{4} f(\bar{x}_i)\cdot \Delta x$$

$$I = 0.5\left[f(\bar{x}_1) + f(\bar{x}_2) + f(\bar{x}_3) + f(\bar{x}_4)\right]$$

$$I = \ldots$$

7. Use the **Composite Simpson's 1/3 Rule** with $n = 4$ intervals to approximate $I = \int_0^2 (x^3 + 1)dx$.

A. 6

B. 6.25

C. 6.125

D. 5.9

$$I = \frac{h}{3}\left[f_0 + 4\sum_{i=1}^{2} f_{(x_{2i-1})} + 2\sum_{i=1}^{1} f_{(x_{2i})} + f_4\right]$$

$$I = \frac{b-a}{4.3}\left[f_0 + 4(f_1 + f_3) + 2f_2 + f_4\right]$$

$$I = \frac{1}{3}\left[f(x_0) + 4f_{(x_1)} + 2f_{(x_2)} + 4f(x_3) + f(x_4)\right]$$

$x_0 = 0,\ x_1 = 0.5,\ x_2 = 1,\ x_3 = 1.5,\ x_4 = 2$

8. Use the **Composite Simpson's 3/8 Rule** with $n = 6$ intervals to approximate $I = \int_0^2 (x^3 + 1)dx$.

A. 6.05

B. 6.25

C. 5.9375

D. 6

$$I = \frac{3h}{8}\left[f_0 + 3\sum_{\substack{i=1 \\ i\not\equiv 0 \pmod 3}}^{5} f_i + 2\sum_{\substack{i=3 \\ i\equiv 0 \pmod 3}}^{3} f_i + f_6\right]$$

$$I = \frac{3\left(\frac{b-a}{6}\right)}{8}\left[f_0 + 3(f_1 + f_2 + f_4 + f_5) + 2f_3 + f_6\right]$$

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9. Which of the following numerical integration rules **requires** the number of intervals (n) to be a multiple of 3?

A. Trapezoidal Rule

B. Midpoint Rule

C. Simpson's $1/3$ Rule

D. Simpson's $3/8$ Rule → this requires 3 intervals

10. Given the Composite Trapezoidal Rule result $T_4 = 6.25$ and the Composite Midpoint Rule result $M_4 = 5.875$ (for $I = \int_0^2 (x^3 + 1)dx$ with $n = 4$, exact value is 6), which of the following is true regarding their absolute error?

A. The Midpoint Rule overestimates the integral, while the Trapezoidal Rule underestimates it.

B. The Trapezoidal Rule error is twice that of the Midpoint Rule error.

C. The Midpoint Rule error is twice that of the Trapezoidal Rule error.

D. Both rules have the same absolute error.

↓ observe first two questions of this quiz-